Tobramycin PopPK Dataset — Quality Control Report

2026-05-14

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Data-Check Summary Table

Check	Description	Result	Status
01	Unique subjects (total count)	300 subjects	ℹ
02	Doses per subject	300 subjects with dose records	ℹ
03a	Distinct dose amounts	49 distinct dose amounts	ℹ
03b	Distinct inter-dose intervals (h)	1 distinct inter-dose interval	ℹ
04	Observed concentration records per subject	300 subjects with concentration records	ℹ
05a	Duplicated sampling time points (EVID=0)	No issues detected	✓
05b	Duplicated concentration records (EVID=0)	No issues detected	✓
06	Negative concentration or dose records	No issues detected	✓
07	Missing concentration records (EVID=0, MDV=1)	No issues detected	✓

Che ck	Description	Result	Stat us
08	Covariate columns with any missing values	No issues detected	✓
09	Time-varying covariates	No issues detected	✓
10	Subjects with non-monotonic TIME	No issues detected	✓
11	Dose records (EVID=1) with missing AMT	No issues detected	✓
12	Obs (EVID=0) with DV present but MDV=1	No issues detected	✓
13	Zero concentration records (EVID=0, MDV=0)	No issues detected	✓
14	Pre-dose observation records	No issues detected	✓
16	Header row quotation marks check	No issues detected	✓

*Legend: **i** = Informational count; ✓ = PASS — no issues detected; ✗ = FAIL — issues flagged.*

Detailed Quality Assessment Report

Dataset Overview

The dataset contains **300 unique subjects** with tobramycin plasma concentration-time profiles recorded over multiple dosing occasions. The dataset includes all structural NONMEM columns (`ID`, `TIME`, `DV`, `AMT`, `EVID`, `MDV`, `CMT`, `RATE`, `DURATION`) as well as a rich set of subject-level covariates, making it suitable for standard two-stage or nonlinear mixed-effects PopPK analysis.

Structural Integrity and Header

The CSV header row is free of quotation marks, and all column names are lowercase and unambiguous — a prerequisite for stable parsing by NONMEM and post-processing tools. The missing data indicator (`.`) is applied consistently across the file for absent `DV` and `AMT` values, in line with NONMEM convention. No structural or formatting issues were identified.

Time Records

`TIME` is monotonically non-decreasing for all 300 subjects. This confirms that records are correctly ordered within each individual’s data block, which is critical for proper event sequencing in NONMEM’s differential equation solver.

Dosing Records

Every subject received exactly **10 doses**, administered as intravenous infusions (`RATE` and `DURATION` columns populated). All doses were given at a uniform inter-dose interval of **24 hours (q24h)**. Dose amounts varied across subjects, ranging from **320 mg to 860 mg** across **49 distinct levels**, a pattern consistent with individualised tobramycin dosing in clinical practice. In routine therapeutic drug monitoring (TDM), tobramycin doses are typically adjusted for body weight and renal function, which explains the wide spread of dose amounts observed.

No dosing records with missing `AMT` were detected. All `EVID = 1` records are structurally complete and correctly populated.

Observation Records

The number of concentration samples per subject ranged from **21 to 28** (median: **25**), indicating a consistent and rich sampling design across the cohort. The following integrity checks all returned zero flagged records:

- No duplicated sampling time points within any individual
- No duplicate concentration records (same `ID`, `TIME`, `DV`)
- No negative observed concentrations
- No zero observed concentrations
- No pre-dose concentration samples (all `EVID = 0` records occur on or after the first dose)
- No records with `MDV = 0` flagged as missing and no observation records where `DV` is present but `MDV = 1` (no contradictory MDV flags)
- No `EVID = 0, MDV = 1` records (no missing-by-design or below-quantification-limit observations flagged in this dataset)

Covariate Assessment

All **8 subject-level covariates** — `age`, `weight`, `sex`, `clcr`, `baseline_scr`, `diabetes`, `hypertension`, and `n_prior_courses` — are complete, with zero missing values across all records and all subjects. All covariates are **static** (time-invariant), which is the expected structure for baseline covariate analysis in a standard PopPK model estimated with NONMEM.

Covariate	Variability	Missing records
age	Static	0
weight	Static	0
sex	Static	0
clcr	Static	0

Covariate	Variability	Missing records
baseline_scr	Static	0
diabetes	Static	0
hypertension	Static	0
n_prior_courses	Static	0

Of particular pharmacokinetic relevance, **creatinine clearance (`clcr`)** and **baseline serum creatinine (`baseline_scr`)** are both fully available. Tobramycin undergoes almost exclusive renal elimination (>95% excreted unchanged in urine), making `clcr` a strong *a priori* candidate covariate for clearance. The availability of baseline `scr` provides an additional renally-sensitive biomarker that may contribute complementary information in covariate model testing.

Recommendations

1. **Prioritise `clcr` in covariate modelling.** Given tobramycin's renal elimination, `clcr` should be the first covariate tested on clearance (`CL`). A power function or linear relationship on the CL_{cr} ratio is a well-established starting point for aminoglycosides.

2. **Investigate the basis for dose individualisation.** The 49 distinct dose amounts (320–860 mg) likely reflect prospective TDM-guided dosing. The dose–`weight` and dose–`clcr` relationships should be explored graphically in the exploratory data analysis to confirm this and to check for any unintended confounding with PK estimates.

3. **Verify intended sampling design.** The slight variation in concentration samples per subject (21–28) is expected in a clinical dataset and does not compromise estimation. A plot of observed sample times relative to dose administration should confirm that peak, mid-interval, and trough samples are captured as intended.

4. **No data cleaning is required.** All integrity checks passed without exception. The dataset can proceed directly to graphical exploratory analysis and PopPK structural model development.

Is this a valid NONMEM dataset?

Yes — this dataset is fully valid for NONMEM analysis. The assessment below confirms that all mandatory and recommended criteria are satisfied.

Criterion	Detail	Status
Essential NONMEM columns present	ID, TIME, DV, AMT, EVID, MDV, CMT, RATE, DURATION all present	✓
Missing data encoded correctly	Period (.) used consistently for missing DV and AMT	✓
Header free of quotation marks	All column headers are clean, lowercase, unquoted	✓
No negative or zero concentrations	None detected across all 300 subjects	✓
TIME monotonically non-decreasing	Confirmed for all 300 subjects	✓
EVID/MDV flags consistent	No contradictory EVID/MDV combinations found	✓
No duplicate records	No duplicated time points or concentration values	✓
All covariate data complete	Zero missing values across all 8 covariates	✓

The tobramycin dataset is **ready for PopPK modelling** and requires no pre-processing prior to NONMEM submission.